Jason Li

Kaysville, UT — 21jason.li@gmail.com — Website — LinkedIn — GitHub

EDUCATION

University of Utah Salt Lake City, UT
Ph.D. in Computing, Data Management and Analysis Track *Aug. 2025 – Present*
- GPA: 4.0/4.0

University of Florida Gainesville, FL
B.S. in Computer Science and Statistics *Aug. 2021 – May 2025*
- GPA: 3.94/4.0; GRE: 337 (Q: 170, V: 167)
- Honors: UF Honors Program, Undergraduate Research Scholars Program, Benacquisto Scholar (Full-Ride Scholarship), Dean's List, National Merit Finalist

PUBLICATIONS

1. **Jason Li**, Rodrigo de la Nuez Moraleda, Dhruv Meduri, Kelin Zia, and Bei Wang, "Explainable Reaction Atlas," *IEEE Workshop on Topology meets Artificial Intelligence (TopoInVis Connect), IEEE VIS*, 2026. **Accepted. Best Paper Award.**

2. Andres Espinosa, Cody Flynn, **Jason Li**, Han Mach, Maximillian Banach, Cathy Quan, Jorg Peters, and Brian Stensrud, "Multi-Agent Board Game Strategy Through Simulation," *Interservice/Industry Training, Simulation and Education Conference (I/ITSEC)*, 2025. **Best Paper Award.**

AWARDS

- **TopoInVis Connect 2026 Best Paper Award** — *Explainable Reaction Atlas* (IEEE VIS)
- **I/ITSEC 2025 Best Paper Award** — *Multi-Agent Board Game Strategy Through Simulation*

CERTIFICATIONS

NVIDIA Certificate of Competency — Building Agentic AI Applications with Large Language Models NVIDIA
Instructor-Led Workshop *Oct. 2026*

RESEARCH EXPERIENCE

Research Assistant Dec. 2025 – Present
Topological Intelligence Lab (TILab), University of Utah *Salt Lake City, UT*
- Conduct research under Ph.D. advisor Professor Bei Wang Phillips at the intersection of explainable AI, deep learning, and data visualization.
- Develop methods for visualizing and interpreting latent representations learned by deep learning models to improve model transparency and understanding.
- Investigate explainable AI techniques and human-AI interaction methods for enabling users to explore, interpret, and reason about complex model behavior.

Research Assistant Feb. 2024 – May 2024
Professor Leo Duan's Research Group, University of Florida *Gainesville, FL*
- Statistically analyzed and visualized travel movement trends caused by Hurricane Ian using R.
- Identified traffic-volume patterns before, during, and after Hurricane Ian.
- Parsed and cleaned census-tract-level geospatial data using Python.

Research Assistant Aug. 2023 – Jan. 2024
Embodied Learning & Experience Lab, University of Florida *Gainesville, FL*
- Developed a smartwatch application to facilitate learning in everyday life for children ages 9–12.
- Reviewed and synthesized peer-reviewed literature to inform research initiatives.

PROFESSIONAL & TEACHING EXPERIENCE

Teaching Assistant Aug. 2024 – May 2025
University of Florida *Gainesville, FL*
- Tutored students in Information and Database Systems.
- Graded student submissions, administered examinations, and presented lecture material during discussion sections.

Software Engineer Intern Jun. 2024 – Aug. 2024
Fidelity Investments *Salt Lake City, Utah*
- Developed a secure web-based funds-transfer platform for digital banking using Angular and GraphQL.
- Implemented backend API services to query and format data and support front-end navigation.

Peer Tutor Oct. 2023 – May 2024
Academic Tutoring Center, University of Florida *Gainesville, FL*
- Provided one-on-one and small-group tutoring in the core programming curriculum for Computer Science and Computer Engineering students.

IT Department Business Analyst Intern May 2022 – Aug. 2022
Supplemental Health Care *Sandy, Utah*
- Automated user-permission assignment, reducing user creation time by 50%.
- Implemented a notification system supporting upskilling for more than 10,000 affiliated nurses.
- Collaborated with developers, stakeholders, project managers, and QA auditors throughout SDLC processes.

PROJECTS

Explainable Reaction Atlas 2026

- Developed an LLM-based explanation framework for interpreting chemical reaction latent spaces represented as interactive tree-structured atlases.
- Implemented Neighborhood and Path Explainers to summarize local reaction regions, compare branches, and analyze transitions between user-selected reactions.
- Evaluated the framework on the Schneider 50K dataset of approximately 50,000 chemical reactions, enabling explanation-guided exploration of reaction patterns and locally inconsistent regions.

Multi-Agent Board Game Strategy Through Simulation 2025

- Developed a custom simulation environment for the turn-based strategy game *Settlers of Catan* to support reinforcement-learning experiments.
- Implemented a Deep Q-Learning agent in PyTorch using an epsilon-greedy exploration strategy.
- Trained and evaluated the agent against a state-of-the-art benchmark, achieving a 64% win rate in one-vs-one simulations.

Household Location Choice Model

- Modeled housing choice behavior in Salt Lake County, Utah, reducing the feature set from 150 variables to 12.
- Applied stepwise regression, LASSO, principal component analysis, random forests, variable transformations, interactions, and genetic-algorithm optimization.

Online Shopping Application

- Built an interactive full-stack shopping application using React, Ant Design, Spring MVC, Hibernate, MySQL, and Node.js.
- Developed REST APIs for registration, search, and checkout and deployed the application using Amazon EC2 and RDS.

Online Room Booking Site

- Developed a React and Spring Boot room-booking application with reservation, search, upload, and deletion services.
- Implemented MySQL persistence, Google Cloud Storage, Elasticsearch geospatial search, and JWT authentication.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, C#, SQL, R, MATLAB, Kotlin

AI/ML & Scientific Computing: PyTorch, TensorFlow, scikit-learn, Hugging Face, NumPy, pandas, Jupyter, RAG, OpenAI API, Model Context Protocol (MCP), Pydantic AI, vLLM

Backend & Data Systems: FastAPI, Node.js, Spring Boot, Spring MVC, .NET Framework, Hibernate, GraphQL, REST APIs, MySQL, Elasticsearch, Apache Tomcat

Frontend, Mobile & Visualization: React, Angular, Svelte, HTML/CSS, D3.js, Android

Cloud & Infrastructure: Docker, Docker Compose, Caddy, AWS EC2, Amazon RDS, Oracle Cloud Infrastructure,

Google App Engine, Google Cloud Storage, Ubuntu/Linux, DNS, HTTPS/TLS

Developer Tools & Practices: Git/GitHub, Pytest, Distributed Computing, Multithreading, Agile (Scrum/Kanban), Jira